

SafarSaarthi — Report

SMART INDIA HACKATHON 2026 | Problem Statement ID: SIH26204 | Team FLUX1

1. Executive Summary

SafarSaarthi is a tourism-safety and trust layer designed for situations where a tourist has to make an unplanned, street-level decision about a driver, guide, route, fare, hotel surroundings, or emergency response. The prototype focuses on turning an uncertain interaction into a more transparent, verifiable and evidence-backed transaction. The current concept combines fare checking, recent passenger payment benchmarks, QR-based identity/credential verification, a digital handshake, safety-hotspot awareness, hotel-area safety, reviews, and SOS/evidence generation.

2. Source Basis and Current Positioning

The supplied SIH presentation identifies the solution as a Travel & Tourism software innovation and describes the core positioning as building trust at the moment of an unbooked tourist interaction. It lists a methodology covering trip personalization, route/destination/stay selection, fair-fare and recent-payment display, QR verification, a recorded digital handshake, review/price feedback, and SOS/evidence generation. The presentation also states that the MVP is highly feasible, while larger-scale deployment depends on verified data, security hardening and local partnerships.

3. Current Prototype — Feature Summary

- **Fare & price intelligence:** Shows fair-rate ranges and recent passenger-payment examples for selected routes; supports quick recognition of potentially unreasonable quoted fares.
- **Digital identity / handshake:** QR-based driver/guide scanning is used to connect identity/credential information with a digital interaction record.
- **Safety hotspot radar:** Map-based hotspot layers and proactive advice can warn users around known risk areas.
- **Hotel-area safety:** The prototype lets a tourist select a hotel and inspect the surrounding area using place/location information.
- **SOS / evidence packet:** The prototype can compile a structured incident dossier containing time, location, scanned-party details, quoted price/benchmark information and other incident evidence.
- **Reviews / trust data:** Post-trip feedback contributes additional trust information for future interactions.

4. Technical Architecture / Stack

- **Frontend:** HTML5, CSS3, JavaScript ES Modules, responsive/mobile-first UI
- **Maps & location:** Google Maps JavaScript API and browser GPS/location capabilities
- **Scanning / OCR:** QRCode.js, html5-qrcode, Tesseract.js
- **Backend / deployment:** Node.js, Firebase Hosting, Cloud Firestore with Firestore Security Rules
- **Prototype persistence:** LocalStorage for prototype profile persistence

5. Current End-to-End Methodology

1. Tourist opens SafarSaarthi and establishes trip context.
2. Trip personalization and route/destination/stay selection are used to localize the experience.
3. Fare intelligence provides a reference range and recent passenger-payment context.
4. Driver/guide QR is scanned and available identity/credential information is presented.
5. A digital handshake is recorded to create a traceable interaction.
6. The trip or service proceeds.
7. Price/review feedback contributes new trust information.
8. If required, SOS/evidence generation creates a structured incident record.

6. Future Implementation Roadmap

- **Verified operator registry:** Partner with tourism departments, transport unions, licensed guides and hotels to establish authoritative operator records. Use signed verification status and expiry dates rather than treating self-entered credentials as authoritative.
- **Real-time fare intelligence:** Move from seeded/reference fare ranges toward regularly refreshed, location-aware benchmarks using authorized tariff sources and anonymized transaction data. Show confidence and data freshness beside every benchmark.
- **Privacy-preserving trust ledger:** Store only the minimum information required for verification, reviews and incident evidence. Separate public reputation signals from personally identifiable information and define retention/deletion rules.
- **Production-grade identity verification:** Introduce stronger onboarding for drivers/guides, including document verification, vehicle/permit checks where legally available, and periodic re-verification.
- **Offline-first travel mode:** Cache essential route, safety and emergency information so core features remain usable with weak or intermittent connectivity.
- **Emergency escalation integrations:** Where formal partnerships exist, integrate emergency workflows with relevant tourist-assistance or police channels. Clearly distinguish an in-app evidence packet from an officially filed complaint.
- **Multilingual and accessibility layer:** Add major Indian and tourist languages, voice guidance, larger-touch targets, high-contrast modes and screen-reader-friendly flows.
- **Fraud / abuse controls:** Detect review manipulation, repeated false reports, QR reuse, suspicious account activity and coordinated rating attacks while providing an appeal/review mechanism.
- **Scalable analytics:** Create dashboards for verified partners showing aggregate hotspot trends, fare anomalies and service-quality signals without exposing unnecessary tourist identities.
- **Pilot deployment:** Start with one destination and a controlled partner group, measure verification coverage, response time, false-alert rate, user completion rate and incident-evidence usefulness before wider rollout.

7. Suggested Phased Roadmap

Phase 1 — Prototype hardening (0–2 months) — Stabilize flows, remove demo-only assumptions, add error states, improve accessibility, formalize data schemas, test security rules and document consent/retention behavior.

Phase 2 — Controlled pilot (2–5 months) — Select one destination, onboard a small set of verified drivers/guides/hotels, validate fare and hotspot data, measure user completion and incident-evidence usefulness.

Phase 3 — Trust infrastructure (5–9 months) — Introduce stronger operator verification, data freshness indicators, privacy-preserving reputation, abuse detection, multilingual support and offline caches.

Phase 4 — Partnership scale-up (9–18 months) — Expand destinations and partner coverage, establish formal tourism/emergency workflows where agreements exist, and introduce aggregate operational dashboards.

8. Risks and Mitigations

- **Data accuracy:** Outdated or incorrect fares/hotspots can mislead users. Mitigation: Show source/freshness/confidence; use verified partners; provide correction workflow.
- **Privacy:** Location, identity and incident evidence are sensitive. Mitigation: Data minimization, access controls, encryption, retention limits and explicit consent.
- **False trust:** A QR scan alone must not imply that an operator is safe. Mitigation: Use precise labels: identity/credential verified, not a blanket safety guarantee.
- **Connectivity:** Tourists may lose network access. Mitigation: Offline cache and graceful degradation for emergency information.
- **Adoption:** Drivers/guides may not initially participate. Mitigation: Partner onboarding, simple QR workflows and incentives for legitimate operators.

9. Success Metrics for a Pilot

- Percentage of target operators with verified profiles.
- Percentage of fare checks that return a current benchmark with a visible freshness indicator.
- Median time from QR scan to verification result.
- Percentage of completed interactions with a recorded digital handshake.
- False-alert / incorrect-hotspot report rate.
- Percentage of incident dossiers successfully generated without missing critical fields.
- User-reported confidence before and after using the relevant feature.
- Partner response time for corrections or verification updates.